Universidad de Sevilla

Estudio de la estructura axial del nucleón en interacciones con neutrinos y antineutrinos

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10 de marzo de 2026

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A. Relaciones cinemáticas e identidades auxiliares

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Cálculo de trazas y contracciones

En este apéndice se recopilan las identidades y pasos intermedios utilizados en el cálculo del tensor leptónico, del tensor hadrónico y de la contracción $\tilde{\eta}_{\mu\nu}\tilde{W}^{\mu\nu}$.

A.1. Álgebra de Clifford y convenciones

Adoptamos unidades naturales ($\hbar = c = 1$) y la convención $\varepsilon^{0123} = +1$. Las matrices gamma satisfacen

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}, \qquad \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3. \tag{A.1}$$

A.2. Trazas básicas

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu] = 4g^{\mu\nu}, \tag{A.2}$$

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = 4\left(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}\right), \tag{A.3}$$

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_5] = -4i\,\varepsilon^{\mu\nu\rho\sigma}, \tag{A.4}$$

$$\begin{aligned}\mathrm{Tr}[\gamma^{\mu_1}\gamma^{\mu_2}\gamma^{\mu_3}\gamma^{\mu_4}\gamma^{\mu_5}\gamma^{\mu_6}] = 4\Big(&g^{\mu_1\mu_2}g^{\mu_3\mu_4}g^{\mu_5\mu_6} - g^{\mu_1\mu_2}g^{\mu_3\mu_5}g^{\mu_4\mu_6} + g^{\mu_1\mu_2}g^{\mu_3\mu_6}g^{\mu_4\mu_5}\\ &- g^{\mu_1\mu_3}g^{\mu_2\mu_4}g^{\mu_5\mu_6} + g^{\mu_1\mu_3}g^{\mu_2\mu_5}g^{\mu_4\mu_6} - g^{\mu_1\mu_3}g^{\mu_2\mu_6}g^{\mu_4\mu_5}\\ &+ g^{\mu_1\mu_4}g^{\mu_2\mu_3}g^{\mu_5\mu_6} - g^{\mu_1\mu_4}g^{\mu_2\mu_5}g^{\mu_3\mu_6} + g^{\mu_1\mu_4}g^{\mu_2\mu_6}g^{\mu_3\mu_5}\\ &- g^{\mu_1\mu_5}g^{\mu_2\mu_3}g^{\mu_4\mu_6} + g^{\mu_1\mu_5}g^{\mu_2\mu_4}g^{\mu_3\mu_6} - g^{\mu_1\mu_5}g^{\mu_2\mu_6}g^{\mu_3\mu_4}\\ &+ g^{\mu_1\mu_6}g^{\mu_2\mu_3}g^{\mu_4\mu_5} - g^{\mu_1\mu_6}g^{\mu_2\mu_4}g^{\mu_3\mu_5} + g^{\mu_1\mu_6}g^{\mu_2\mu_5}g^{\mu_3\mu_4}\Big).\end{aligned} \tag{A.5}$$

y la traza de un número impar de matrices gamma es nula.

A.3. Proyectores quirales

Usaremos $P_L = \frac{1-\gamma_5}{2}$, $P_R = \frac{1+\gamma_5}{2}$, de modo que

$$(1-\gamma_5)^2 = 2(1-\gamma_5). \tag{A.6}$$

A.4. Cálculo del tensor leptónico

Introducimos la notación

$$\tilde{\gamma}_\mu \equiv \gamma_\mu(1-\gamma_5) = \tilde{\gamma}_\mu^V + \tilde{\gamma}_\mu^A, \tag{A.7}$$

y utilizamos la abreviatura $\not{k} \equiv \gamma_\alpha k^\alpha$. Para el cálculo por trazas (aplicando completitud de espinores), aparece el objeto

$$\mathcal{T}_{\mu\nu} \;=\; \mathrm{Tr}\left[\frac{\not{k}_f + m_\mu}{2m_\mu}\,\tilde{\gamma}_\mu\,\frac{\not{k}_i + m_\nu}{2m_\nu}\,\tilde{\gamma}_\nu\right], \tag{A.8}$$

donde k_i y k_f denotan los cuadrimomentos inicial (del neutrino) y final (del leptón cargado), y m_ν y m_μ sus respectivas masas.

Sustituyendo $\tilde{\gamma}_\mu = \gamma_\mu(1-\gamma_5)$ y $\tilde{\gamma}_\nu = \gamma_\nu(1-\gamma_5)$, y desarrollando los paréntesis, se obtiene

$$\begin{aligned}
\mathcal{T}_{\mu\nu} &= \mathrm{Tr}\left[\frac{\not{k}_f + m_\mu}{2m_\mu}\,\gamma_\mu(1-\gamma_5)\,\frac{\not{k}_i + m_\nu}{2m_\nu}\,\gamma_\nu(1-\gamma_5)\right] \\
&= \frac{1}{(2m_\mu)(2m_\nu)}\,\mathrm{Tr}\big[(\not{k}_f\gamma_\mu(1-\gamma_5) + m_\mu\gamma_\mu(1-\gamma_5))\,(\not{k}_i\gamma_\nu(1-\gamma_5) + m_\nu\gamma_\nu(1-\gamma_5))\big] \\
&= \frac{1}{4m_\mu m_\nu}\,\mathrm{Tr}\Big[\not{k}_f\gamma_\mu(1-\gamma_5)\not{k}_i\gamma_\nu(1-\gamma_5) + \not{k}_f\gamma_\mu(1-\gamma_5)\,m_\nu\gamma_\nu(1-\gamma_5) \\
&\qquad + m_\mu\gamma_\mu(1-\gamma_5)\not{k}_i\gamma_\nu(1-\gamma_5) + m_\mu\gamma_\mu(1-\gamma_5)\,m_\nu\gamma_\nu(1-\gamma_5)\Big]. \qquad \text{(A.9)}
\end{aligned}$$

A partir de aquí, separamos la contribución total en cuatro trazas:

$$\mathcal{T}_{\mu\nu} = \frac{1}{4m_\mu m_\nu}\left(T^{(1)}_{\mu\nu} + T^{(2)}_{\mu\nu} + T^{(3)}_{\mu\nu} + T^{(4)}_{\mu\nu}\right), \qquad \text{(A.10)}$$

con

$$T^{(1)}_{\mu\nu} \equiv \mathrm{Tr}\big[\not{k}_f\gamma_\mu(1-\gamma_5)\not{k}_i\gamma_\nu(1-\gamma_5)\big]\,, \qquad \text{(A.11)}$$

$$T^{(2)}_{\mu\nu} \equiv m_\nu\ \mathrm{Tr}\big[\not{k}_f\gamma_\mu(1-\gamma_5)\gamma_\nu(1-\gamma_5)\big]\,, \qquad \text{(A.12)}$$

$$T^{(3)}_{\mu\nu} \equiv m_\mu\ \mathrm{Tr}[\gamma_\mu(1-\gamma_5)\not{k}_i\gamma_\nu(1-\gamma_5)]\,, \qquad \text{(A.13)}$$

$$T^{(4)}_{\mu\nu} \equiv m_\mu m_\nu\ \mathrm{Tr}[\gamma_\mu(1-\gamma_5)\gamma_\nu(1-\gamma_5)]\,. \qquad \text{(A.14)}$$

Simplificación usando γ_5

La expansión de los factores quirales puede reorganizarse usando las propiedades

$$\{\gamma_5, \gamma^\alpha\} = 0, \qquad \gamma_5^2 = \mathbb{I}, \qquad \gamma_5\,\not{a}\,\gamma_5 = -\not{a}, \qquad \gamma_5\,\gamma^\alpha\,\gamma_5 = -\gamma^\alpha, \qquad \text{(A.15)}$$

válidas para cualquier cuadrivector a^μ. En particular,

$$P_L\,\gamma^\alpha\,P_L = 0, \qquad P_L = \frac{1-\gamma_5}{2}, \qquad \text{(A.16)}$$

lo cual explica por qué ciertos términos con dos proyectores consecutivos se anulan.

Consideremos el núcleo matricial que aparece en las trazas de (B.9),

$$\gamma_\mu(1-\gamma_5)\,\not{k}_i\,\gamma_\nu(1-\gamma_5). \qquad \text{(A.17)}$$

Al expandir $(1-\gamma_5)(1-\gamma_5)$ se obtiene

$$\gamma_\mu(1-\gamma_5)\,\not{k}_i\,\gamma_\nu(1-\gamma_5) = \gamma_\mu\not{k}_i\gamma_\nu - \gamma_\mu\gamma_5\not{k}_i\gamma_\nu - \gamma_\mu\not{k}_i\gamma_\nu\gamma_5 + \gamma_\mu\gamma_5\not{k}_i\gamma_\nu\gamma_5. \qquad \text{(A.18)}$$

El último término puede simplificarse aplicando (**??**):

$$\gamma_\mu\gamma_5\not{k}_i\gamma_\nu\gamma_5 = \gamma_\mu(\gamma_5\not{k}_i\gamma_5)(\gamma_5\gamma_\nu\gamma_5) = \gamma_\mu(-\not{k}_i)(-\gamma_\nu) = \gamma_\mu\not{k}_i\gamma_\nu. \qquad \text{(A.19)}$$

Por tanto, en cualquier traza en la que aparezca el bloque anterior, los términos "(1)" y "(4)" son iguales, y la combinación completa se reduce a

$$\mathrm{Tr}[\cdots\gamma_\mu(1-\gamma_5)\not{k}_i\gamma_\nu(1-\gamma_5)\cdots] = 2\ \mathrm{Tr}[\cdots\gamma_\mu\not{k}_i\gamma_\nu\cdots] - \mathrm{Tr}[\cdots\gamma_\mu\gamma_5\not{k}_i\gamma_\nu\cdots] - \mathrm{Tr}[\cdots\gamma_\mu\not{k}_i\gamma_\nu\gamma_5\cdots]. \qquad \text{(A.20)}$$

Aplicando esta identidad a la descomposición de (B.10), puede tomarse directamente

$$T^{(4)}_{\mu\nu} = T^{(1)}_{\mu\nu} \qquad \text{(A.21)}$$

de modo que la suma de cuatro trazas de la expresión (B.18) se reescribe como

$$\gamma_\mu\not{k}_i\gamma_\nu - \gamma_\mu\gamma_5\not{k}_i\gamma_\nu - \gamma_\mu\not{k}_i\gamma_\nu\gamma_5 + \gamma_\mu\gamma_5\not{k}_i\gamma_\nu\gamma_5 = 2T^{(A)}_{\mu\nu} - T^{(B)}_{\mu\nu} - T^{(C)}_{\mu\nu} \qquad \text{(A.22)}$$

Evaluación de $T^{(A)}_{\mu\nu}$, $T^{(B)}_{\mu\nu}$ y $T^{(C)}_{\mu\nu}$

Tras la simplificación (véase la discusión previa), basta evaluar

$$T_{\mu\nu} = 2T^{(A)}_{\mu\nu} - T^{(B)}_{\mu\nu} - T^{(C)}_{\mu\nu}. \tag{A.23}$$

Usamos $\not{k} \equiv \gamma_\alpha k^\alpha$, y denotamos por k_i y k_f los cuadrimomentos inicial y final del leptón. Recordamos que la traza de un número impar de matrices gamma es nula.

(A) Término vectorial. Definimos

$$T^{(A)}_{\mu\nu} \equiv \mathrm{Tr}\big[\not{k}_f\, \gamma_\mu\, \not{k}_i\, \gamma_\nu\big] = k_f^\alpha k_i^\beta\ \mathrm{Tr}[\gamma_\alpha\gamma_\mu\gamma_\beta\gamma_\nu]\,. \tag{A.24}$$

Usando $\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$, se obtiene

$$T^{(A)}_{\mu\nu} = 4\left(k_{f\mu}k_{i\nu} + k_{f\nu}k_{i\mu} - (k_f\cdot k_i)\, g_{\mu\nu}\right). \tag{A.25}$$

(B) Término axial (con γ_5). Definimos

$$T^{(B)}_{\mu\nu} \equiv \mathrm{Tr}\big[\not{k}_f\, \gamma_\mu\, \not{k}_i\, \gamma_\nu\, \gamma_5\big] = k_f^\alpha k_i^\beta\ \mathrm{Tr}[\gamma_\alpha\gamma_\mu\gamma_\beta\gamma_\nu\gamma_5]\,. \tag{A.26}$$

Usando $\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_5] = -4i\,\varepsilon^{\mu\nu\rho\sigma}$, obtenemos

$$T^{(B)}_{\mu\nu} = -4i\,\varepsilon_{\alpha\mu\beta\nu}\, k_f^\alpha k_i^\beta. \tag{A.27}$$

(C) Segundo término axial (orden equivalente). De forma análoga,

$$T^{(C)}_{\mu\nu} \equiv \mathrm{Tr}\big[\not{k}_f\, \gamma_\mu\, \not{k}_i\, \gamma_\nu\, \gamma_5\big] = -4i\,\varepsilon_{\alpha\mu\beta\nu}\, k_f^\alpha k_i^\beta, \tag{A.28}$$

de manera que $T^{(B)}_{\mu\nu} = T^{(C)}_{\mu\nu}$ en este caso.

Resultado final. Sustituyendo (**??**)–(**??**) en (**??**) se obtiene

$$T_{\mu\nu} = 8\left(k_{f\mu}k_{i\nu} + k_{f\nu}k_{i\mu} - (k_f\cdot k_i)\, g_{\mu\nu}\right) + 8i\,\varepsilon_{\alpha\mu\beta\nu}\, k_f^\alpha k_i^\beta. \tag{A.29}$$

A.5. Contracciones de tensores de Levi-Civita

Se emplea la identidad

$$\varepsilon_{\alpha\mu\beta\nu}\,\varepsilon^{\mu\nu\rho\lambda} = -2\Big(\delta_\alpha{}^\rho\delta_\beta{}^\lambda - \delta_\alpha{}^\lambda\delta_\beta{}^\rho\Big), \tag{A.30}$$

y, en particular,

$$A^\alpha B^\beta\,\varepsilon_{\alpha\mu\beta\nu}\,\varepsilon^{\mu\nu\rho\lambda} = -2\left(A^\rho B^\lambda - A^\lambda B^\rho\right). \tag{A.31}$$

A.6. Tensor hadrónico: descomposición vectorial, axial y vector–axial

Partimos de la definición (promedio sobre el espín inicial y suma sobre el final)

$$\begin{aligned}\tilde{W}^{\mu\nu} &\equiv \sum_{s_f,s_i} \Big[\bar{u}_p(p_f,s_f)\,\tilde{\Gamma}^{\mu}\,u_n(p_i,s_i)\Big]\,\Big[\bar{u}_p(p_f,s_f)\,\overline{\tilde{\Gamma}^{\nu}}\,u_n(p_i,s_i)\Big] \\ &= \frac{1}{8M^2}\,\mathrm{Tr}\Big[(\not{p}_f+M)\,\tilde{\Gamma}^{\mu}\,(\not{p}_i+M)\,\overline{\tilde{\Gamma}^{\nu}}\Big]\,. \end{aligned} \tag{A.32}$$

En el régimen cuasi-elástico, el vértice hadrónico efectivo se parametriza como

$$\tilde{\Gamma}^{\mu} = F_1^V\,\gamma^{\mu} + i\,\frac{F_2^V}{2M}\,\sigma^{\mu\alpha}Q_{\alpha} + G_A\,\gamma^{\mu}\gamma^5 + F_P\,Q^{\mu}\gamma^5, \tag{A.33}$$

$$\overline{\tilde{\Gamma}^{\nu}} = F_1^V\,\gamma^{\nu} - i\,\frac{F_2^V}{2M}\,\sigma^{\nu\beta}Q_{\beta} + G_A\,\gamma^{\nu}\gamma^5 - F_P\,Q^{\nu}\gamma^5, \tag{A.34}$$

donde $\sigma^{\mu\nu} \equiv \frac{i}{2}[\gamma^{\mu},\gamma^{\nu}]$.

Para separar contribuciones vectoriales y axiales, definimos

$$\tilde{\Gamma}^{\mu} = \tilde{\Gamma}^{\mu}_V + \tilde{\Gamma}^{\mu}_A, \qquad \tilde{\Gamma}^{\mu}_V \equiv F_1^V\,\gamma^{\mu} + i\,\frac{F_2^V}{2M}\,\sigma^{\mu\alpha}Q_{\alpha}, \qquad \tilde{\Gamma}^{\mu}_A \equiv G_A\,\gamma^{\mu}\gamma^5 + F_P\,Q^{\mu}\gamma^5. \tag{A.35}$$

Sustituyendo (**??**) en (**??**) y expandiendo,

$$\begin{aligned}\tilde{W}^{\mu\nu} &= \frac{1}{8M^2}\,\mathrm{Tr}\Big[(\not{p}_f+M)\,(\tilde{\Gamma}^{\mu}_V+\tilde{\Gamma}^{\mu}_A)\,(\not{p}_i+M)\,(\overline{\tilde{\Gamma}^{\nu}_V}+\overline{\tilde{\Gamma}^{\nu}_A}\Big] \\ &= \frac{1}{8M^2}\,\mathrm{Tr}\Big[(\not{p}_f+M)\,\tilde{\Gamma}^{\mu}_V\,(\not{p}_i+M)\,\overline{\tilde{\Gamma}^{\nu}_V}\Big] + \frac{1}{8M^2}\,\mathrm{Tr}\Big[(\not{p}_f+M)\,\tilde{\Gamma}^{\mu}_A\,(\not{p}_i+M)\,\overline{\tilde{\Gamma}^{\nu}_A}\Big] \\ &\quad + \frac{1}{8M^2}\,\mathrm{Tr}\Big[(\not{p}_f+M)\,\tilde{\Gamma}^{\mu}_V\,(\not{p}_i+M)\,\overline{\tilde{\Gamma}^{\nu}_A} + (\not{p}_f+M)\,\tilde{\Gamma}^{\mu}_A\,(\not{p}_i+M)\,\overline{\tilde{\Gamma}^{\nu}_V}\Big] \\ &\equiv \tilde{W}^{\mu\nu}_V + \tilde{W}^{\mu\nu}_A + \tilde{W}^{\mu\nu}_{VA}. \end{aligned} \tag{A.36}$$

A.6.1. Término vectorial: descomposición en $f_{11}^{\mu\nu}$, $f_{22}^{\mu\nu}$ y $f_{12}^{\mu\nu}$

El término vectorial del tensor hadrónico se escribe como

$$\tilde{W}^{\mu\nu}_V = \frac{1}{8M^2}\,\mathrm{Tr}\Big[(\not{p}_f+M)\,\tilde{\Gamma}^{\mu}_V\,(\not{p}_i+M)\,\overline{\tilde{\Gamma}^{\nu}_V}\Big]\,. \tag{A.37}$$

Sustituyendo

$$\tilde{\Gamma}^{\mu}_V = F_1^V\,\gamma^{\mu} + i\,\frac{F_2^V}{2M}\,\sigma^{\mu\alpha}Q_{\alpha}, \qquad \overline{\tilde{\Gamma}^{\nu}_V} = F_1^V\,\gamma^{\nu} - i\,\frac{F_2^V}{2M}\,\sigma^{\nu\beta}Q_{\beta}, \tag{A.38}$$

obtenemos

$$\tilde{W}^{\mu\nu}_V = \frac{1}{8M^2}\,\mathrm{Tr}\Big[(\not{p}_f+M)\left(F_1^V\gamma^{\mu} + i\frac{F_2^V}{2M}\sigma^{\mu\alpha}Q_{\alpha}\right)(\not{p}_i+M)\left(F_1^V\gamma^{\nu} - i\frac{F_2^V}{2M}\sigma^{\nu\beta}Q_{\beta}\right)\Big]\,. \tag{A.39}$$

Desarrollando el producto se obtiene (nótese que el factor $(\not{p}_i + M)$ aparece en ambos términos cruzados):

$$\begin{aligned}
\tilde{W}_V^{\mu\nu} &= \frac{1}{8M^2}\ \mathrm{Tr}\Big[(\not{p}_f + M)\,(F_1^V\gamma^\mu)\,(\not{p}_i + M)\,(F_1^V\gamma^\nu)\Big] \\
&\quad + \frac{1}{8M^2}\ \mathrm{Tr}\left[\left(i\frac{F_2^V}{2M}\right)\left(-i\frac{F_2^V}{2M}\right)(\not{p}_f + M)\,\sigma^{\mu\alpha}Q_\alpha\,(\not{p}_i + M)\,\sigma^{\nu\beta}Q_\beta\right] \\
&\quad + \frac{1}{8M^2}\ \mathrm{Tr}\left[(\not{p}_f + M)\left(i\frac{F_2^V}{2M}\sigma^{\mu\alpha}Q_\alpha\right)(\not{p}_i + M)(F_1^V\gamma^\nu) - (\not{p}_f + M)(F_1^V\gamma^\mu)(\not{p}_i + M)\left(i\frac{F_2^V}{2M}\sigma^{\nu\beta}Q_\beta\right)\right. \\
&= \frac{1}{8M^2}\left[(F_1^V)^2\,f_{11}^{\mu\nu} + \frac{(F_2^V)^2}{4M^2}\,f_{22}^{\mu\nu} + \frac{i\,F_1^V F_2^V}{2M}\,f_{12}^{\mu\nu}\right], \qquad \text{(A.40)}
\end{aligned}$$

donde definimos

$$f_{11}^{\mu\nu} \equiv \mathrm{Tr}\Big[(\not{p}_f + M)\,\gamma^\mu\,(\not{p}_i + M)\,\gamma^\nu\Big], \qquad \text{(A.41)}$$

$$f_{22}^{\mu\nu} \equiv \mathrm{Tr}\Big[(\not{p}_f + M)\,\sigma^{\mu\alpha}Q_\alpha\,(\not{p}_i + M)\,\sigma^{\nu\beta}Q_\beta\Big], \qquad \text{(A.42)}$$

$$f_{12}^{\mu\nu} \equiv \mathrm{Tr}\Big[(\not{p}_f + M)\,\sigma^{\mu\alpha}Q_\alpha\,(\not{p}_i + M)\,\gamma^\nu - (\not{p}_f + M)\,\gamma^\mu\,(\not{p}_i + M)\,\sigma^{\nu\beta}Q_\beta\Big]. \qquad \text{(A.43)}$$

Cálculo de $f_{11}^{\mu\nu}$

Calculamos $f_{11}^{\mu\nu}$ paso a paso:

$$\begin{aligned}
f_{11}^{\mu\nu} &= \mathrm{Tr}\Big[(\not{p}_f + M)\,\gamma^\mu\,(\not{p}_i + M)\,\gamma^\nu\Big] \\
&= \mathrm{Tr}\Big[\not{p}_f\,\gamma^\mu\,\not{p}_i\,\gamma^\nu\Big] + M\ \mathrm{Tr}\Big[\not{p}_f\,\gamma^\mu\,\gamma^\nu\Big] + M\ \mathrm{Tr}\Big[\gamma^\mu\,\not{p}_i\,\gamma^\nu\Big] + M^2\ \mathrm{Tr}[\gamma^\mu\gamma^\nu]. \qquad \text{(A.44)}
\end{aligned}$$

Los dos términos intermedios se anulan por contener un número impar de matrices gamma, $\mathrm{Tr}[\gamma\gamma\gamma] = 0$. Por tanto,

$$\begin{aligned}
f_{11}^{\mu\nu} &= \mathrm{Tr}\Big[\not{p}_f\,\gamma^\mu\,\not{p}_i\,\gamma^\nu\Big] + M^2\ \mathrm{Tr}[\gamma^\mu\gamma^\nu] \\
&= p_f^\rho p_i^\sigma\ \mathrm{Tr}[\gamma_\rho\gamma^\mu\gamma_\sigma\gamma^\nu] + 4M^2\,g^{\mu\nu}. \qquad \text{(A.45)}
\end{aligned}$$

Usando

$$\mathrm{Tr}[\gamma^\rho\gamma^\mu\gamma^\sigma\gamma^\nu] = 4\,(g^{\rho\mu}g^{\sigma\nu} - g^{\rho\sigma}g^{\mu\nu} + g^{\rho\nu}g^{\mu\sigma}), \qquad \text{(A.46)}$$

se obtiene

$$\begin{aligned}
p_f^\rho p_i^\sigma\ \mathrm{Tr}[\gamma_\rho\gamma^\mu\gamma_\sigma\gamma^\nu] &= 4\,p_f^\rho p_i^\sigma\,\left(g_\rho{}^\mu g_\sigma{}^\nu - g_{\rho\sigma}g^{\mu\nu} + g_\rho{}^\nu g_\sigma{}^\mu\right) \\
&= 4\left(p_f^\mu p_i^\nu + p_f^\nu p_i^\mu - (p_f\cdot p_i)\,g^{\mu\nu}\right). \qquad \text{(A.47)}
\end{aligned}$$

Finalmente,

$$f_{11}^{\mu\nu} = 4\left[p_f^\mu p_i^\nu + p_f^\nu p_i^\mu + \left(M^2 - p_f\cdot p_i\right)g^{\mu\nu}\right]. \qquad \text{(A.48)}$$

Cálculo de $f_{22}^{\mu\nu}$

Partimos de

$$f_{22}^{\mu\nu} \equiv \mathrm{Tr}\Big[(\not{p}_f + M)\,\sigma^{\mu\alpha}Q_\alpha\,(\not{p}_i + M)\,\sigma^{\nu\beta}Q_\beta\Big]. \qquad \text{(A.49)}$$

1) Expansión en masas y cancelación de trazas impares. Desarrollando $(\not{p}_{f,i}+M)$,

$$f_{22}^{\mu\nu} = \mathrm{Tr}\Big[\not{p}_f\, \sigma^{\mu\alpha} Q_\alpha\, \not{p}_i\, \sigma^{\nu\beta} Q_\beta\Big] + M\ \mathrm{Tr}\Big[\sigma^{\mu\alpha} Q_\alpha\, \not{p}_i\, \sigma^{\nu\beta} Q_\beta\Big]$$
$$+ M\ \mathrm{Tr}\Big[\not{p}_f\, \sigma^{\mu\alpha} Q_\alpha\, \sigma^{\nu\beta} Q_\beta\Big] + M^2\ \mathrm{Tr}\big[\sigma^{\mu\alpha} Q_\alpha\, \sigma^{\nu\beta} Q_\beta\big]\,. \tag{A.50}$$

Los términos lineales en M contienen un número impar de matrices gamma y se anulan, por lo que

$$f_{22}^{\mu\nu} = \mathrm{Tr}\Big[\not{p}_f\, \sigma^{\mu\alpha} Q_\alpha\, \not{p}_i\, \sigma^{\nu\beta} Q_\beta\Big] + M^2\ \mathrm{Tr}\big[\sigma^{\mu\alpha} Q_\alpha\, \sigma^{\nu\beta} Q_\beta\big]\,. \tag{A.51}$$

2) Término M^2: traza $\sigma\sigma$. Usamos

$$\mathrm{Tr}\big(\sigma^{AB}\sigma^{CD}\big) = 4\left(g^{AC}g^{BD} - g^{AD}g^{BC}\right), \tag{A.52}$$

de modo que, para $A=\mu, B=\alpha, C=\nu, D=\beta$,

$$\mathrm{Tr}\big[\sigma^{\mu\alpha} Q_\alpha\, \sigma^{\nu\beta} Q_\beta\big] = 4\left(g^{\mu\nu}Q^2 - Q^\mu Q^\nu\right). \tag{A.53}$$